

TABLE XXXII

D13 NATURAL CLEAN POSITIVE-TRANSFER AUDIT. THE TARGET IS TO BEAT BOTH SAR AND CONFIDENCE-WEIGHTED MEAN (CW) WITHOUT SYNTHETIC DEGRADATION. OPENED DEVELOPMENT ROWS CANNOT SET GATE E; OFFICIAL ROWS STILL REQUIRE BOTH CO-ENDPOINTS TO PASS.

Dataset	Rule	Δ vs SAR (95% CI)	Δ vs CW (95% CI)	Status
3D-ADAM	residual stack	+0.0583 [+0.0410, +0.0759]	+0.0182 [+0.0075, +0.0290]	opened-dev positive
MulSen	residual stack	+0.0374 [-0.0824, +0.1575]	+0.0193 [-0.0488, +0.0883]	opened-dev unresolved
Real-IAD	residual stack	-0.0858 [-0.0928, -0.0787]	+0.0191 [+0.0148, +0.0235]	official not confirmed

TABLE XXXIII

D16 REAL-IAD D3 NATURAL-DEGRADATION AUDIT. THE PRIMARY ENDPOINT IS THE STRESS SUBSET VERSUS CONFIDENCE-WEIGHTED MEAN (CW); ALL-TEST ROWS ARE SUPPORTIVE ONLY.

Endpoint	Comparator	Δ AUC (95% CI)	Decision
Stress subset	CW	+0.0351 [-0.0276, +0.0980]	fail: CI crosses zero
Clean subset	CW	+0.0000 [+0.0000, +0.0000]	pass: no regression
All test	CW	+0.0624 [+0.0347, +0.0896]	supportive only
All test	SAR	-0.2939 [-0.3213, -0.2665]	fail

TABLE XXXIV

STRESS-REGIME TRANSFER ON HELD-OUT EXTERNAL 3D-ADAM: DEPTH DEGRADED BY α . RGA'S KS-DRIFT GATE SIGNIFICANTLY BEATS CONFIDENCE-WEIGHTING (CW) ONCE A MODALITY DEGRADES ($\alpha \geq 0.5$), WHILE TIES ON CLEAN DATA.

α	r_{depth}	Static	Conf.-wt.	RGA	RGA-CW
0.00	0.586	0.9349	0.9353	0.9349	-0.0004
0.25	0.034	0.8845	0.8848	0.8847	-0.0001
0.50	0.019	0.8730	0.8732	0.8769	+0.0036*
0.75	0.016	0.8348	0.8351	0.8733	+0.0382*
1.00	0.019	0.7471	0.7474	0.8667	+0.1194*

gate_s_natural_degradation_confirmed=false; D16 is supportive natural-degradation evidence, not a gate pass.

g) *Role of the new Real-IAD D3 data.*: The new Real-IAD D3 data is used as a natural-degradation audit, not as a retuning set and not as a replacement definition for Gate E. Its purpose is to check whether the stress-regime mechanism observed under controlled degradation also produces measurable headroom on real degraded samples. The answer is mixed: the data reveals a positive full-test signal against CW, but also shows that the prelocked stress-subset endpoint and the stronger SAR comparison still fail. Thus the dataset improves the evidence boundary and failure diagnosis, while leaving flagship readiness unchanged.

h) *Scope of the competitive-detector results.*: We state three boundaries explicitly. First, the patch-level PatchCore upstream detector was applied to the two naturally paired RGB+depth benchmarks analysed here (MVTec 3D-AD and the held-out external 3D-ADAM); the remaining benchmark cells (VisA, MVTec LOCO-AD, UNSW-NB15, Real3D-AD) retain the diagnostic upstream detector and were *not* re-run, so their numbers should be read under that detector, not the competitive one. Second, the in-domain 0.904/0.978 ROC-AUC figures are obtained under the *supervised-paired* protocol, which exposes positives to the fusion training fold; they are therefore an in-domain strong-baseline comparison and are *not*

comparable to the published one-class MVTec 3D-AD leaderboard, whose evaluation we leave to future work. Third, the stress-regime transfer result uses a *controlled synthetic* degradation (a uniform-noise blend) that isolates the differential-reliability mechanism; demonstrating the same crossover under naturally occurring sensor degradation is future work. None of these boundaries is load-bearing for the claim actually made—a multi-split-confirmed in-domain win and a replicated stress-regime advantage—but each bounds how far that claim may be read.

J. Leakage and Independence Sanity Checks

Two reviewer criticisms deserve direct sanity-check experiments rather than prose-only defences.

UNSW-NB15 held-out-attack-category protocol. UNSW-NB15 is known to share attack categories between train and test, so the headline 0.989 ROC-AUC could in principle be explained by within-attack leakage rather than genuine cross-attack generalisation. We add a held-out-attack-category variant using `src/scripts/prepare_unsw_paired_fusion_benchmark.py` with Worms, Shellcode, and Backdoor held out. This routes those three attack categories *entirely* into the test fold, leaving the fusion train and validation folds free of those attacks. The fusion model is graded on its ability to flag attack categories it never saw during training. Under this protocol, RGA+ router reaches 0.994 ROC-AUC, still beating every non-router baseline (random forest 0.993, MLP 0.992, Tent 0.955, TTT 0.960, late-fusion ensemble 0.944). The held-out result is materially *higher* than the unrestricted UNSW supervised paired result because the held-out attacks are individually easier to separate from normal traffic than the original mix; what matters is that the RGA+ delta against its validation-frozen primary comparator is preserved. The descriptive 0.989 / 0.993 numbers therefore do not depend on within-attack leakage; this is reported as an exploratory Family C cell rather than as a generalization claim.

Edge-proxy independence noise-floor. For VisA and MVTec LOCO-AD the second "domain" is a Sobel-edge proxy derived deterministically from the same RGB image, which is a real limitation: the two domains are not independent. To quantify how much of the headline supervised-paired result depends on edge_proxy carrying genuine signal (rather than RGA+ effectively pulling on the RGB stream alone), we build a noise-floor variant of VisA SP that replaces the edge_proxy score and embedding columns with uniform random samples (`src/scripts/derive_noise_floor_csv.py`). Un-